

Linear Regression Analysis

Henrik Navarroza

July 6

```
##A
library(terra)

## terra 1.9.34

library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2

## -- Conflicts ----- tidyverse_conflicts() --
## x tidyr::extract() masks terra::extract()
## x dplyr::filter()  masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

setwd("C:/Users/Admin/downloads")
herpedata <- read.csv("herpa.csv")
herpeclean <- herpedata[complete.cases(herpedata$SPPTotal), ]
phil_dem <- rast("PHL_msk_alt.tif")
herpecoord <- herpeclean[, c("Long", "Lat")]
coord <- vect(as.matrix(herpecoord), crs = crs(phil_dem))
coordelev <- terra::extract(phil_dem, coord)

richelev <- data.frame(Richness = herpeclean$SPPTotal, Elevation = coordelev[,2])
head(richelev)

##   Richness Elevation
## 1         0        284
## 2         0        284
## 3         0        124
## 4         0        124
## 5         1        564
## 6         1        564
```

```

x <- richlev$Elevation
y <- richlev$Richness
xbar <- mean(x)
ybar <- mean(y)
b <- sum((x - xbar) * (y - ybar)) / sum((x - xbar)^2)
a <- ybar - b * xbar

cat("Regression Equation:\n")

```

Regression Equation:

```

cat("y=", round(a,3), "+", round(b,3), "x\n")

```

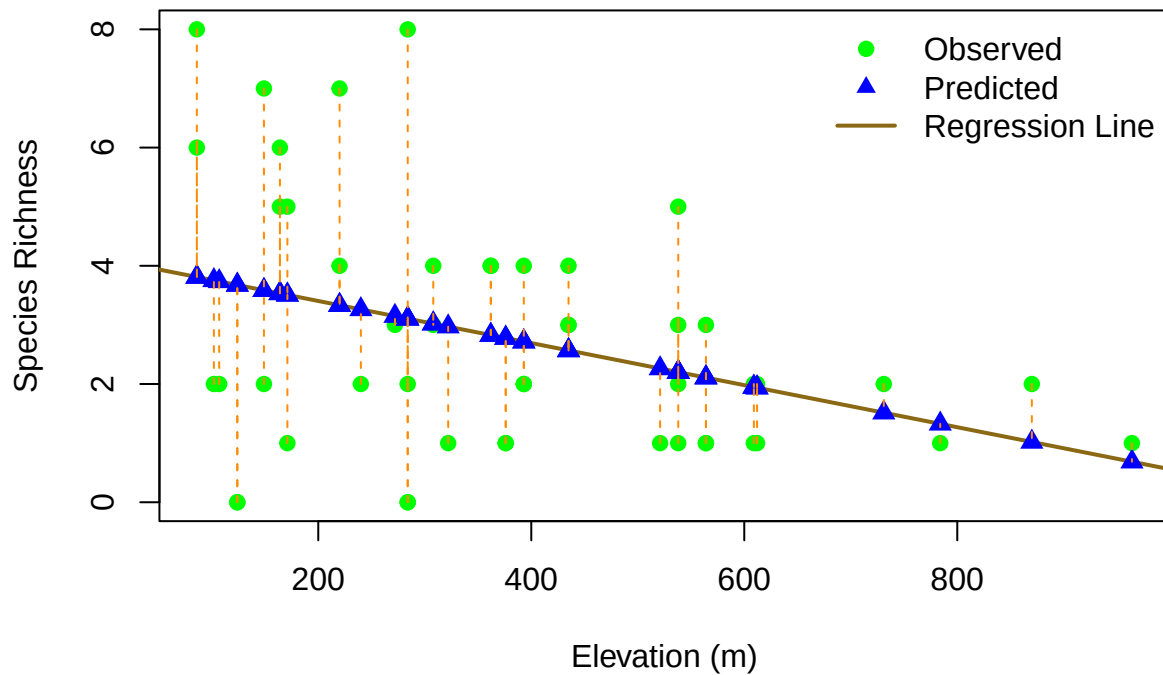
$y = 4.116 + -0.004 x$

```

yhat <- a + b * x
plot(x, y,
     pch = 19, col = "green",
     xlab = "Elevation (m)", ylab = "Species Richness",
     main = "Simple Linear Regression Model")
abline(a, b, col = "goldenrod4", lwd = 2)
points(x, yhat, pch = 17, col = "blue", cex = 1.2)
segments(x0 = x, y0 = y, x1 = x, y1 = yhat, lty = 2, col = "darkorange")
legend("topright",
     legend = c("Observed", "Predicted", "Regression Line"),
     pch = c(19, 17, NA),
     lty = c(NA, NA, 1),
     col = c("green", "blue", "goldenrod4"),
     lwd = c(NA, NA, 2),
     bty = "n")

```

Simple Linear Regression Model



Insights

The downward-sloping regression line indicates a statistically significant negative relationship between

#B

```
phil_rain <- rast("annual_precipitation.tif")
phil_temp <- rast("annual_temperature.tif")
coordrain <- terra::extract(phil_rain, coord)
coordtemp <- terra::extract(phil_temp, coord)

combined <- data.frame(
  Richness = herpeclean$SPPTotal,
  Elevation = coordelev[,2],
  Precipitation = coordrain[,2],
  Temperature = coordtemp[,2])

model <- lm(Richness ~ Elevation + Precipitation + Temperature, data = combined)

x <- combined$Elevation
x1 <- combined$`Precipitation`
x2 <- combined$`Temperature`
y <- combined$Richness
model <- lm(y ~ x + x1 + x2)
coef(model)
```

```
## (Intercept)          x          x1          x2
## 41.83631964  0.01427313 -0.04211388  1.06904218
```

```

yhat <- predict(model)
residuals <- residuals(model)
results <- data.frame(
  Elevation = x,
  "Annual Precipitation" = x1,
  "Annual Temperature" = x2,
  Observed = y,
  Predicted = round(yhat,2),
  Residual = round(residuals,2)
)
print(results)

```

##	Elevation	Annual.Precipitation	Annual.Temperature	Observed	Predicted
## 1	284	1714	26.23333	0	1.75
## 2	284	1714	26.23333	0	1.75
## 3	124	1678	26.98750	0	1.79
## 4	124	1678	26.98750	0	1.79
## 5	564	1788	24.94583	1	1.25
## 6	564	1788	24.94583	1	1.25
## 7	521	1771	25.11667	1	1.54
## 8	964	1883	22.57500	1	0.43
## 9	612	1780	24.18333	1	1.46
## 10	376	1684	25.43750	1	3.48
## 11	376	1684	25.43750	1	3.48
## 12	538	1734	24.69167	1	2.89
## 13	609	1754	24.42500	1	2.77
## 14	784	1803	23.60000	1	2.32
## 15	171	1637	26.33333	1	3.49
## 16	322	1680	25.92917	1	3.40
## 17	284	1714	26.23333	2	1.75
## 18	240	1698	26.42917	2	2.01
## 19	870	1877	23.55417	2	0.39
## 20	731	1810	23.66250	2	1.34
## 21	612	1780	24.18333	2	1.46
## 22	393	1710	25.21250	2	2.38
## 23	393	1710	25.21250	2	2.38
## 24	538	1734	24.69167	2	2.89
## 25	609	1754	24.42500	2	2.77
## 26	102	1619	26.67917	2	3.63
## 27	149	1637	26.79167	2	3.66
## 28	107	1630	26.97500	2	3.56
## 29	564	1788	24.94583	3	1.25
## 30	435	1704	25.15000	3	3.17
## 31	435	1704	25.15000	3	3.17
## 32	538	1734	24.69167	3	2.89
## 33	308	1672	25.69583	3	3.29
## 34	272	1667	26.15833	3	3.48
## 35	393	1710	25.21250	4	2.38
## 36	435	1704	25.15000	4	3.17
## 37	308	1672	25.69583	4	3.29
## 38	220	1651	26.17083	4	3.42
## 39	362	1692	25.65417	4	3.17
## 40	362	1692	25.65417	4	3.17

## 41	538	1734	24.69167	5	2.89
## 42	171	1637	26.33333	5	3.49
## 43	164	1598	26.49167	5	5.20
## 44	86	1579	26.81250	6	5.23
## 45	164	1598	26.49167	6	5.20
## 46	220	1651	26.17083	7	3.42
## 47	149	1637	26.79167	7	3.66
## 48	284	1714	26.23333	8	1.75
## 49	86	1579	26.81250	8	5.23
##	Residual				
## 1	-1.75				
## 2	-1.75				
## 3	-1.79				
## 4	-1.79				
## 5	-0.25				
## 6	-0.25				
## 7	-0.54				
## 8	0.57				
## 9	-0.46				
## 10	-2.48				
## 11	-2.48				
## 12	-1.89				
## 13	-1.77				
## 14	-1.32				
## 15	-2.49				
## 16	-2.40				
## 17	0.25				
## 18	-0.01				
## 19	1.61				
## 20	0.66				
## 21	0.54				
## 22	-0.38				
## 23	-0.38				
## 24	-0.89				
## 25	-0.77				
## 26	-1.63				
## 27	-1.66				
## 28	-1.56				
## 29	1.75				
## 30	-0.17				
## 31	-0.17				
## 32	0.11				
## 33	-0.29				
## 34	-0.48				
## 35	1.62				
## 36	0.83				
## 37	0.71				
## 38	0.58				
## 39	0.83				
## 40	0.83				
## 41	2.11				
## 42	1.51				
## 43	-0.20				
## 44	0.77				

```
## 45      0.80
## 46      3.58
## 47      3.34
## 48      6.25
## 49      2.77
```

```
summary(model)
```

```
##
## Call:
## lm(formula = y ~ x + x1 + x2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.4880 -1.5553 -0.1998  0.8002  6.2488
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 41.83632   39.90717   1.048  0.30008
## x           0.01427    0.01273   1.122  0.26798
## x1          -0.04211    0.01501  -2.806  0.00738 **
## x2           1.06904    1.90400   0.561  0.57726
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.806 on 45 degrees of freedom
## Multiple R-squared:  0.305, Adjusted R-squared:  0.2586
## F-statistic: 6.581 on 3 and 45 DF, p-value: 0.0008767
```

```
# Insights
```

```
#The multiple linear regression model indicates that annual precipitation is the only statistically sig
```

```
““
```